

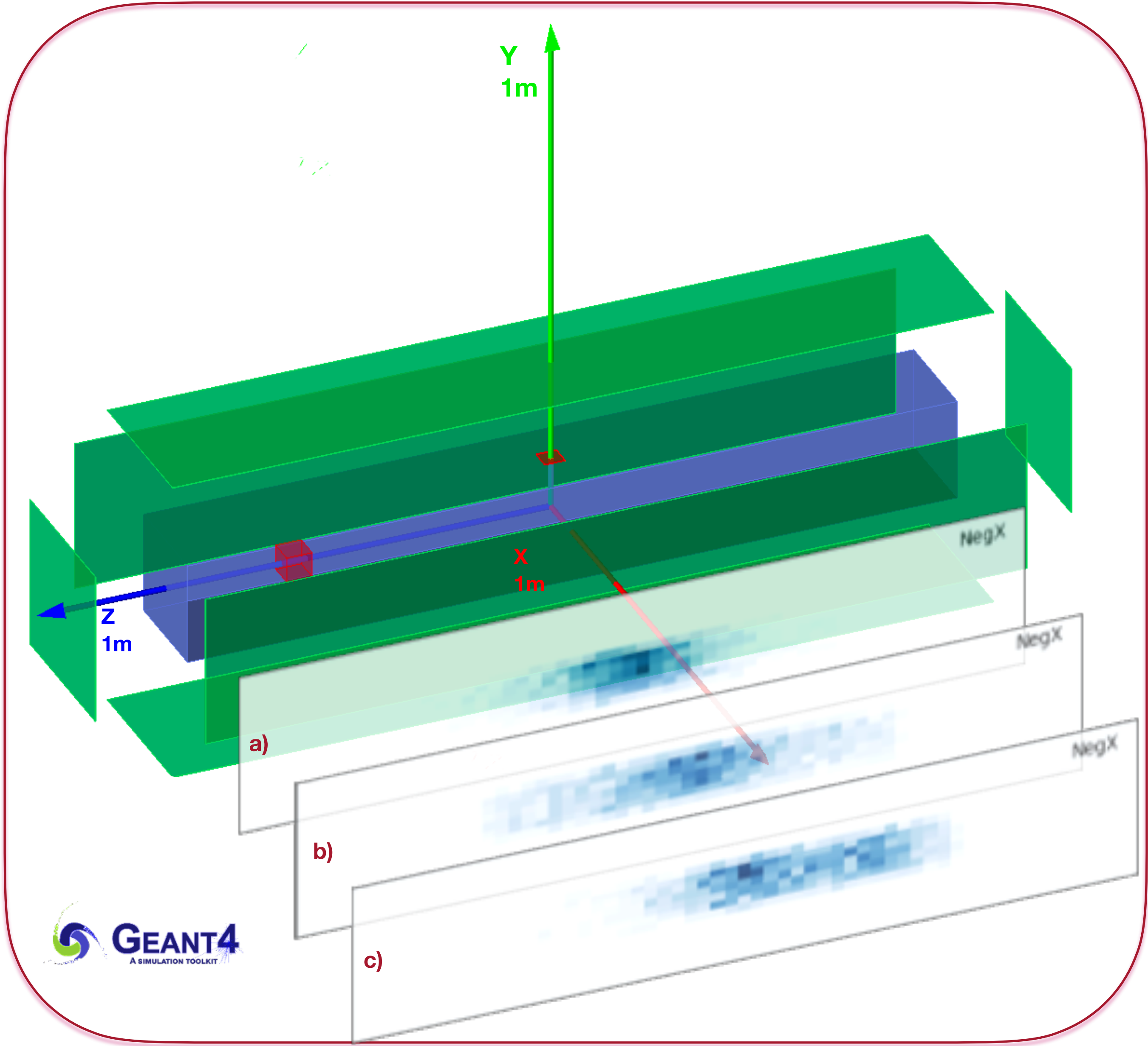
Gamma detection for source localisation in targeted alpha therapy

Motivation

Targeted alpha therapies deliver highly localised dose, but treatment success depends on where the radionuclide source remains after administration. In AlphaGlue delivery [1], alpha emitters are applied to a tumour region in a thin glue layer. Source material may remain at the intended site, diffuse into surrounding tissue, or redistribute via the blood stream. We test whether decay-chain gamma emissions can monitor source position and redistribution on clinically relevant timescales.

Geometry and display

Gamma entries on six detector panels are used to image source position. The geometry shows the AlphaGlue source, water phantom, detector panels, and example display projections for retained (a), diffuse(b), and localised (c) source states.



Key result: Total detector counts changed only weakly, while spatial gamma-hit profiles separated retained, diffuse, and localised source configurations.

Simulation Model

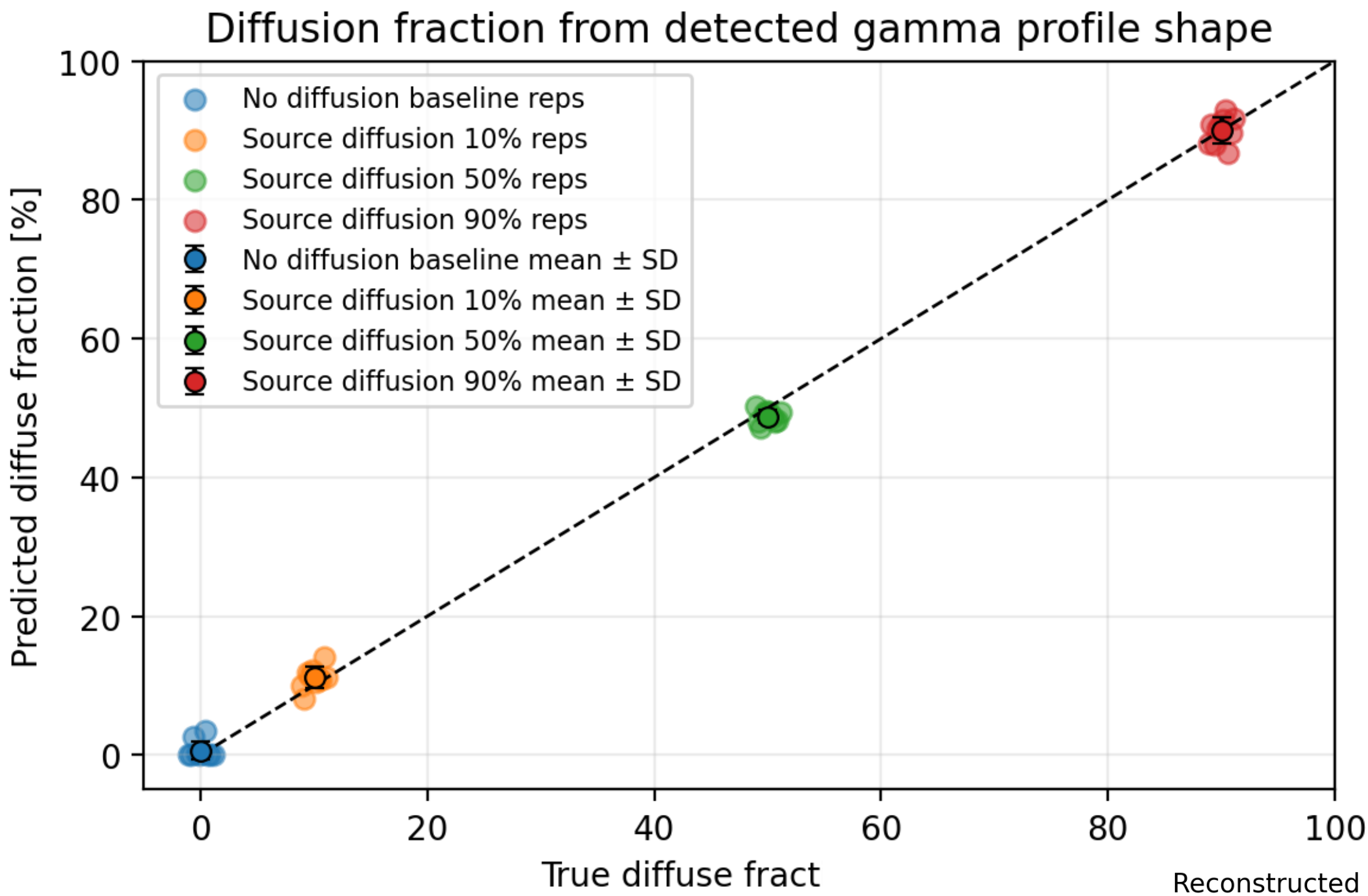
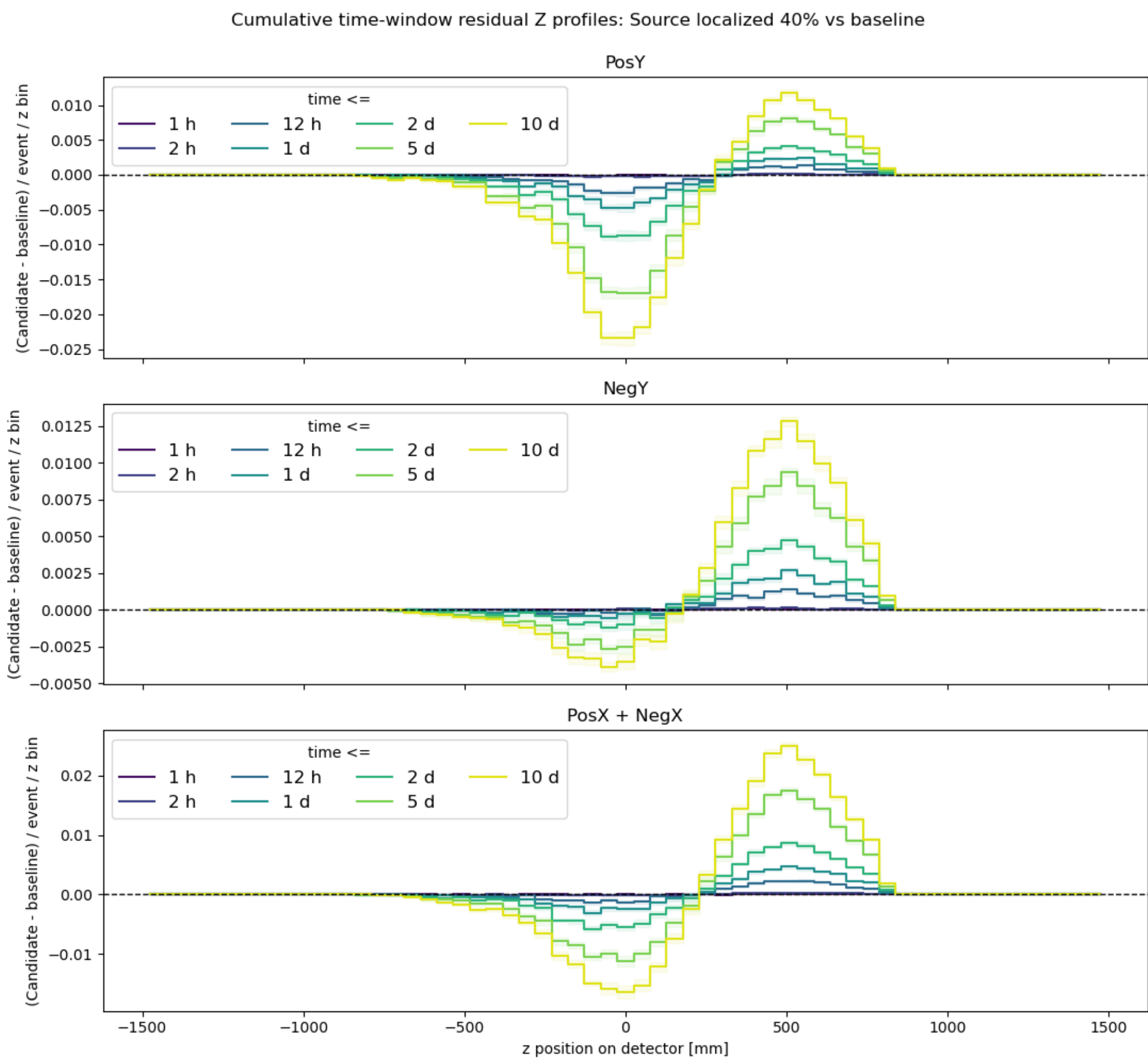
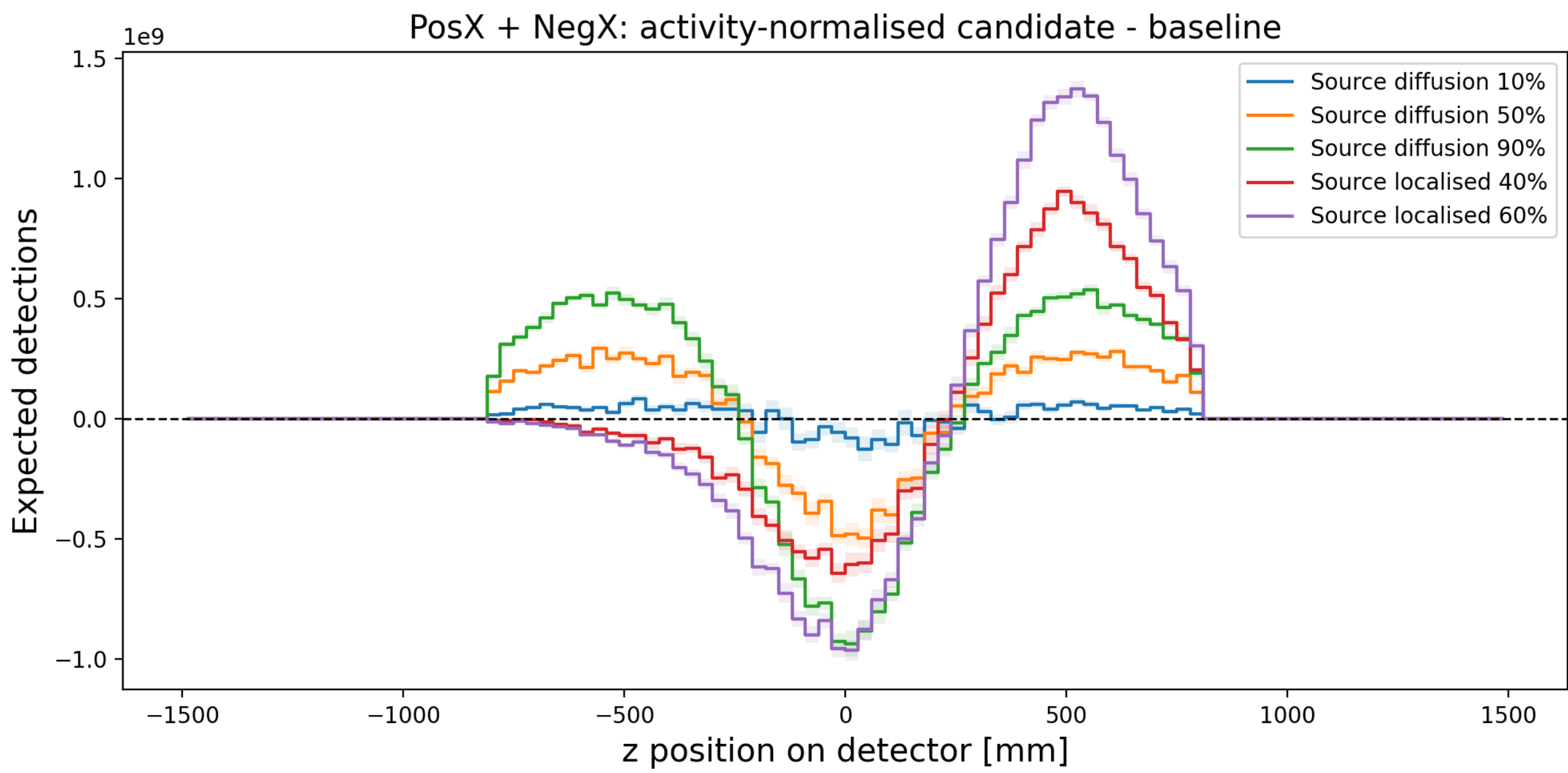
A Geant4 model [4] was used to simulate treatment delivery and gamma detection. The geometry included a water phantom, six photon-detector panels, and a thin AlphaGlue source on the phantom surface. Simulated source states represented retained activity, broad diffusion, and localised displacement, with varying activities.

Source States

CASE	SOURCE STATE
Baseline	100% retained in glue
Diff10/ 50 / 90	10 / 50 / 90% broadly diffused
Loc20/ 40/ 60	20 / 40 / 60% localised displacement

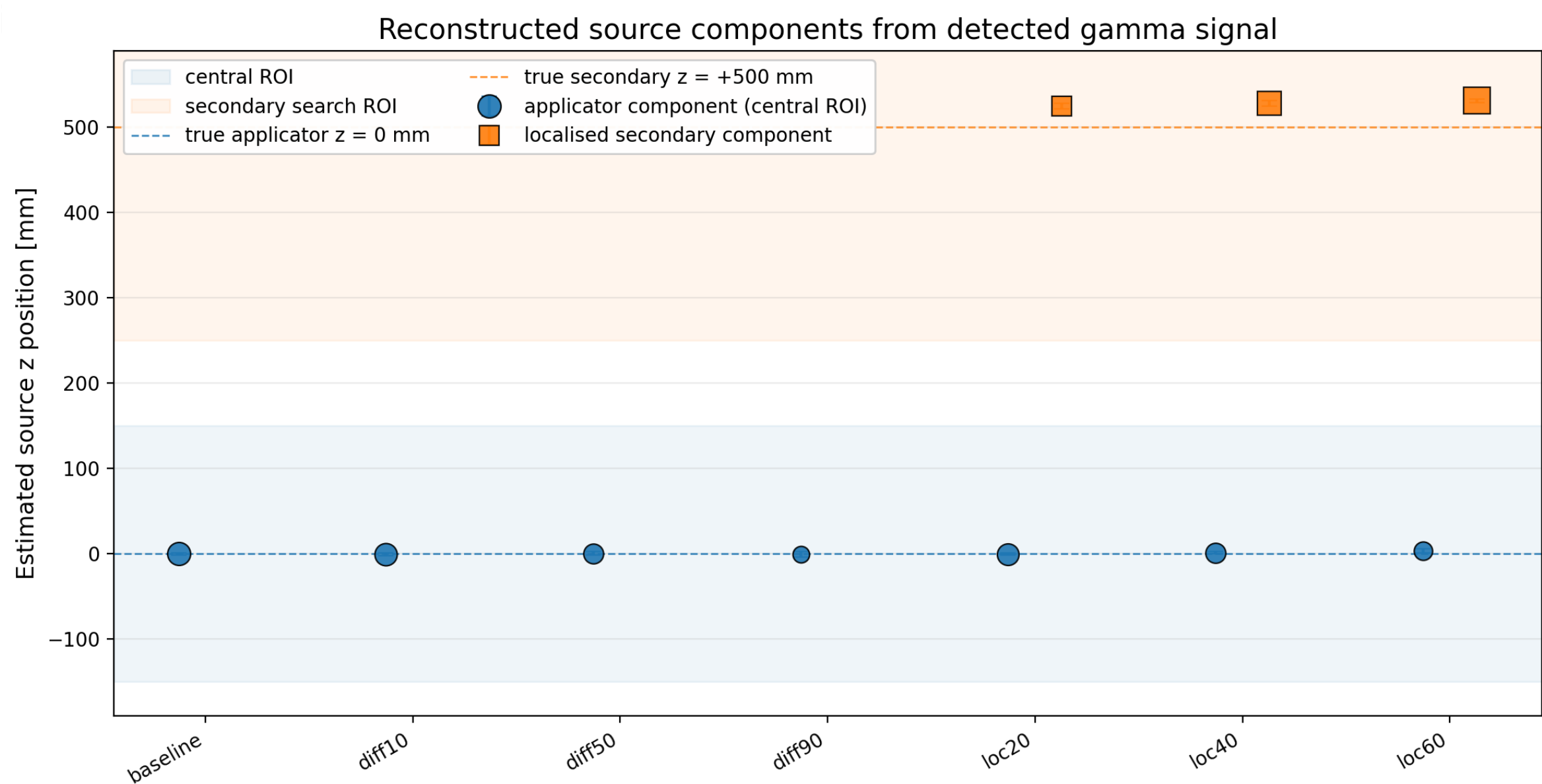
Baseline-subtracted profiles

Candidate detector profiles were baseline-subtracted against the retained-source case. Redistribution produces structured residuals: diffuse activity broadens the response, while localised displacement gives a spatially shifted excess.



Diffusion and localisation

Profile-shape features were extracted from baseline-subtracted detector signals across ten repetitions of 10,000 primary decays. Broad diffusion fraction was recovered with $R^2 \approx 0.998$. Localised displaced-source cases produced a separate position signature.



Time to detection

Cumulative time-window profiles show when redistribution becomes detectable. Larger redistributions crossed threshold within 30 minutes to 2 hours; weaker movement required longer integration.

References

[1] L. Abu Sabah et al., “AlphaGlue: A Novel Conceptual Delivery Method for α Therapy,” *BioMedInformatics* 5, 58, 2025.
[2] L. Ballisat et al., “Geant4 simulation of the enhanced in-vivo ^{220}Rn spread in DaRT,” *Phys. Med.* 135, 105005, 2025.
[3] G. Heger et al., “First measurements of radon-220 diffusion in mice tumors,” *Med. Phys.* 51, 5045–5058, 2024.
[4] S. Agostinelli et al., “Geant4—a simulation toolkit,” *Nucl. Instrum. Methods Phys. Res. A* 506, 250–303, 2003.
[5] L. Arazi et al., “The treatment of solid tumors by alpha emitters released from ^{224}Ra -loaded sources,” *Phys. Med. Biol.* 55, 1203, 2010.
[6] S. Incerti et al., “The Geant4-DNA project,” *Int. J. Model. Simul. Sci. Comput.* 1, 157–178, 2010.

Conclusions

Gamma detection provides source-localisation information beyond total count rate alone. This could support post-treatment monitoring by identifying retained, diffused, and locally displaced activity after AlphaGlue delivery.